Roll No....................

MID SEMESTER EXAMINATION 2025

Name of the Program: **B.Tech. (Electrical Engineering)**
Semester:3^{rds}
Name of the Course: **Mathematics III** Course Code: TMA 301
Time: **90 Minutes** Maximum Marks: **50**

Note:

(i) Answer **all the questions** by choosing *any one of the sub questions*.
(ii) Each question carries 10 marks.

Q1	(10 marks)	
(a)	A card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting (i) a red card, (ii) (ii) a king.	CO1
	OR	
(b)	The probability of a girl passing in an exam is 3/5, and a boy passing in an exam is 4/5. Find the probability that at least one of them has passed the exam?	
Q2	(10 marks)	
(a)	Find the probability of arranging 8 different books if 6 books are always together?	CO1
	OR	
(b)	A factory produces bulbs. The probability that a bulb is defective is 0.02. Find the probability that in a sample of 4 bulbs, (i) none is defective, (ii) exactly one is defective.	
Q3	(10 marks)	
(a)	Find the binomial distribution of getting a six in three tosses of an unbiased dice.	CO1
	OR	
(b)	Find the probability of getting at least 5 times head on tossing an unbiased coin for 6 times by using the binomial distribution.	
Q4	(10 marks)	
(a)	Define a random variable with an example and describe its type.	CO2
	OR	
(b)	If a fair coin is tossed 8 times, find the probability of (i) Exactly 5 heads (ii) At least 5 heads	
Q5	(10 marks)	
(a)	If there are three machines, A, B, and C, producing 25%, 35%, and 40% of the total bolts, then the probabilities of defective bolts by A, B, and C are 5%, 4%, and 2% of the total bolts. A bolt is drawn at random from the product. If the drawn bolt is found to be defective, what is the probability	CO2

	that it is manufactured by (i) Machine A (ii) Machine B (iii) Machine C
	OR
(b)	For probability distribution given, <table><tr><td>X</td><td>-2</td><td>3</td><td>1</td></tr><tr><td>P(X)</td><td>1/3</td><td>1/2</td><td>1/6</td></tr></table>(i) E(X) (ii) $E(X^2)$ (iii) Var (X) (iv) E(2X+5)